

Session 5 — Analysis

Solutions

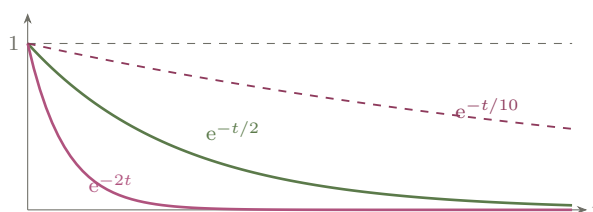
Each answer gives the reasoning, not only the result. Three numbers are used throughout and nothing else is needed: $e^{-1} \approx 0.37$, $e^{-2} \approx 0.14$, $e^{-3} \approx 0.05$. Where a rule has a condition attached, the condition is stated, because almost every mistake in this subject consists of applying a rule outside its conditions.

Functions and the exponential

Exercise 1 *Growth and decay. (a) Does e^{3t} grow or decay? And e^{-3t} , and $e^{-t/5}$? (b) Write $e^{-t/5}$ as e^{at} and give a and τ . (c) Order $e^{-t/2}$, $e^{-t/10}$, e^{-2t} from the fastest decay to the slowest.*

- (a) Write each one as e^{at} and read the sign of a ; nothing else is needed. e^{3t} has $a = 3 > 0$ and **grows**. e^{-3t} has $a = -3 < 0$ and **decays**. $e^{-t/5}$ has $a = -1/5 < 0$ and **decays**.
- (b) $e^{-t/5} = e^{at}$ with $a = -\frac{1}{5}$. Since $a = -1/\tau$ we get $\tau = -1/a = 5$.
- (c) The three time constants are $\tau = 2$, $\tau = 10$ and $\tau = \frac{1}{2}$. A *smaller* τ means a faster decay, so the order is

$$e^{-2t} (\tau = \frac{1}{2}) \succ e^{-t/2} (\tau = 2) \succ e^{-t/10} (\tau = 10).$$



The three curves of part (c), all starting at 1. The pink curve, with the smallest time constant, is the one that falls first; the dashed wine curve, with the largest, has hardly begun to fall over the same interval.

Exercise 2 *Recall decays as $R(t) = R_0 e^{-t/\tau}$ with $R_0 = 0.8$ and $\tau = 3$ days. (a) Compute $R(0)$, $R(3)$, $R(6)$, $R(9)$. (b) Sketch the curve. (c) Someone says “after 3 days the memory is gone”. Why is that wrong?*

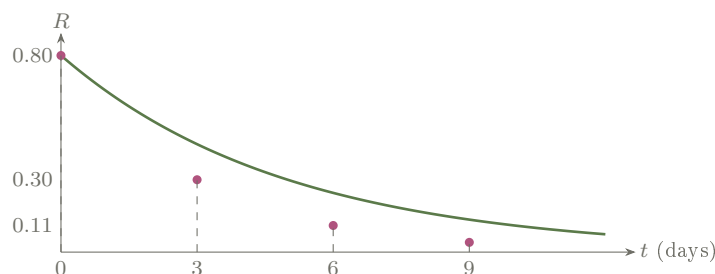
- (a) The delays 0, 3, 6, 9 are 0, 1, 2, 3 time constants, so the factors are 1, 0.37, 0.14, 0.05:

$$R(0) = 0.8, \quad R(3) \approx 0.30, \quad R(6) \approx 0.11, \quad R(9) \approx 0.04.$$

No calculator is needed: only R_0 and the three memorised numbers.

- (b) See the figure. The curve starts at 0.8, falls steeply, then flattens; it approaches the horizontal axis without ever meeting it, because $e^{-t/\tau} > 0$ for every t .

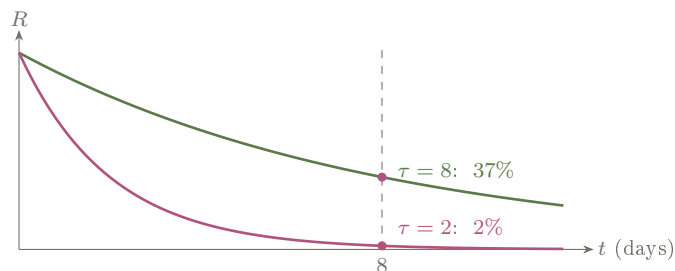
- (c) After 3 days 37% of what was recalled at $t = 0$ is still recalled, which is not nothing. The time constant is the time needed to fall *to* 37%, not the time at which the quantity reaches zero. It measures a rate, not an end.



$R(t) = 0.8e^{-t/3}$. The four marked delays are 0, one, two and three time constants. Each step of 3 days to the right multiplies the height by 0.37 again, which is why the curve flattens rather than reaching zero.

Exercise 3 Two participants forget with $\tau = 2$ days and $\tau = 8$ days, both from $R_0 = 1$. (a) After 8 days, how much does each recall? (b) Which has the better memory, and is that the larger or the smaller τ ?

- (a) Count time constants, then apply the factor 0.37 once for each. For $\tau = 8$, eight days is exactly one time constant, so about **37%** remains. For $\tau = 2$, eight days is four time constants, so $e^{-4} \approx 0.018$, about **2%**.
- (b) The participant with $\tau = 8$. **A larger τ means slower forgetting.** This is worth saying aloud, because the intuition often runs the other way: τ is a duration, and a longer duration means the memory survives longer.



Both curves start at 1. At $t = 8$ days the green curve has completed one time constant and the pink curve four. The vertical dashed line is the single moment at which the two are compared.

Derivatives

Exercise 4 Differentiate, naming the rule each time: (a) $f(t) = t^2 - 4t + 7$; (b) $g(t) = e^{3t}$; (c) $h(t) = 5e^{-2t}$; (d) $k(t) = 1/t + \sqrt{t}$.

- (a) Linearity, then the power rule on each term: $t^2 \rightarrow 2t$, $-4t \rightarrow -4$, $7 \rightarrow 0$ (a constant has zero slope). Hence $f'(t) = 2t - 4$.
- (b) The exponential rule $e^{at} \rightarrow ae^{at}$ with $a = 3$: $g'(t) = 3e^{3t}$.

- (c) The same rule with $a = -2$; the constant factor 5 is carried along unchanged: $h'(t) = 5 \times (-2)e^{-2t} = -10e^{-2t}$.
- (d) The power rule again, this time with a negative and a fractional exponent. Write $k(t) = t^{-1} + t^{1/2}$; then

$$k'(t) = -1 \cdot t^{-2} + \frac{1}{2}t^{-1/2} = -\frac{1}{t^2} + \frac{1}{2\sqrt{t}}.$$

The rule $t^n \rightarrow nt^{n-1}$ holds for every real n , not only for positive integers; that is the whole content of (d).

Exercise 5 *Base or exponent? (a) Differentiate t^2 and e^{2t} : which rule for which, and why? (b) Someone writes $\frac{d}{dt}e^{-t/2} = (-t/2)e^{(-t/2)-1}$. What is wrong, and what is the correct answer?*

- (a) In t^2 the variable is the *base* and the exponent is a fixed number, so the power rule applies: $t^2 \rightarrow 2t$. In e^{2t} the base is the fixed number e and the variable sits in the *exponent*, so the exponential rule applies: $e^{2t} \rightarrow 2e^{2t}$. The two rules are not interchangeable, and the position of t is what selects between them.
- (b) They applied the power rule to an exponential: they treated the exponent $-t/2$ as if it were the constant n of $t^n \rightarrow nt^{n-1}$. The correct answer uses the exponential rule with $a = -\frac{1}{2}$:

$$\frac{d}{dt}e^{-t/2} = -\frac{1}{2}e^{-t/2}.$$

variable in the **base** $\Rightarrow$ power rule

$$t^2 \longrightarrow 2t$$

variable in the **exponent** $\Rightarrow$ exponential rule

$$e^{2t} \longrightarrow 2e^{2t}$$

The single question to ask before differentiating anything of this shape: where does t stand? An answer of the form $(-t/2)e^{(-t/2)-1}$ is the signature of having asked it the wrong way round.

Exercise 6 *(a) Differentiate $R(t) = 0.8e^{-t/4}$ and check that $R' = -R/4$. (b) Say in words what $R' = -R/4$ asserts. (c) Differentiate $P(t) = 100 - 60e^{-t/5}$: is P rising or falling?*

- (a) Here $a = -\frac{1}{4}$, so $R'(t) = -\frac{1}{4} \times 0.8e^{-t/4} = -\frac{1}{4}R(t)$. ✓ The check is not decoration: an exponential is the only shape for which the derivative comes back as a multiple of the function itself, so seeing that multiple appear confirms both the rule and the arithmetic.
- (b) The rate at which recall falls is proportional to how much is still recalled. A lot is lost early, when there is a lot to lose, and less and less later. Read the other way round, this is a *differential equation*: the rate is expressed in terms of the quantity, which is exactly the situation of the last section.
- (c) The constant 100 differentiates to 0, and the exponential brings down $a = -\frac{1}{5}$:

$$P'(t) = -60 \times \left(-\frac{1}{5}\right)e^{-t/5} = +12e^{-t/5}.$$

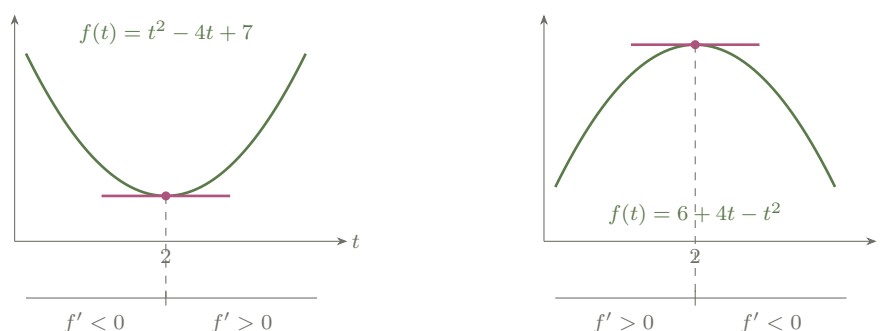
Two negative numbers were multiplied, so the result is positive; and an exponential is never zero, so $P' > 0$ for every t : P is **rising**.

Watch out — the usual mistake

Part (c) is where the sign is lost every year. The minus sign in front of the 60 and the minus sign inside the exponent are two different things, and they multiply. If in doubt, look at the picture: P starts at 40 and climbs towards 100, so its derivative cannot be negative.

Exercise 7 Using the sign of the derivative. (a) $f(t) = t^2 - 4t + 7$: where is $f' = 0$, and is it a maximum or a minimum? (b) $f(t) = 6 + 4t - t^2$: same questions. (c) $P(t) = 100 - 60e^{-t/5}$: show $P' > 0$ everywhere and say what P approaches.

- (a) $f'(t) = 2t - 4$, which vanishes at $t = 2$. To the left of 2, $f' < 0$; to the right, $f' > 0$. Decreasing then increasing is a **minimum**, and $f(2) = 3$.
- (b) $f'(t) = 4 - 2t$, which vanishes at $t = 2$ as well. Now $f' > 0$ before and $f' < 0$ after: increasing then decreasing is a **maximum**, and $f(2) = 10$. The two functions share the location of the stationary point and differ in its nature; only the sign on either side distinguishes them.
- (c) $P'(t) = 12e^{-t/5}$. An exponential is strictly positive for every t , so $P' > 0$ everywhere and P increases without interruption. As t grows, $e^{-t/5} \rightarrow 0$ and $P(t) \rightarrow 100$: the constant term is the value that P approaches, and never attains.



Parts (a) and (b). In both, f' vanishes at $t = 2$ and the pink tangent there is horizontal. What separates a minimum from a maximum is not the stationary point itself but the sign of f' on either side of it, shown on the line below each curve.

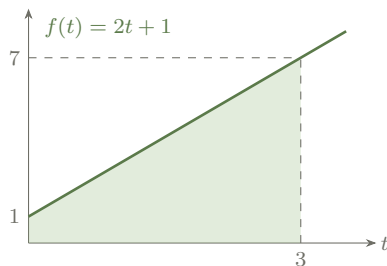
Integrals

Exercise 8 Compute, and check each answer by differentiating it: (a) $\int_0^3 (2t+1) dt$; (b) $\int_0^2 (3t^2 - 4t + 1) dt$; (c) $\int_0^\infty e^{-2t} dt$; (d) an antiderivative of $e^{-t/5}$.

Every definite integral here follows the same three steps: find an antiderivative F of the function written inside the integral, evaluate F at the upper limit, subtract F at the lower limit. The constant of integration cancels in the subtraction, so any antiderivative may be used.

- (a) $F(t) = t^2 + t$, so $\int_0^3 (2t+1) dt = F(3) - F(0) = (9+3) - 0 = \mathbf{12}$. Check: $F'(t) = 2t+1$. ✓
- (b) $F(t) = t^3 - 2t^2 + t$, so the integral is $F(2) - F(0) = (8-8+2) - 0 = \mathbf{2}$. Check: $F'(t) = 3t^2 - 4t + 1$. ✓

- (c) $\int e^{at} dt = e^{at}/a$ with $a = -2$, so $F(t) = -\frac{1}{2}e^{-2t}$. As $t \rightarrow \infty$, $e^{-2t} \rightarrow 0$ and $F(t) \rightarrow 0$; at the lower limit $F(0) = -\frac{1}{2}$. The integral is $0 - (-\frac{1}{2}) = \frac{1}{2}$. This is the time constant of e^{-2t} , as the next exercise explains.
- (d) $F(t) = -5e^{-t/5}$. Check: $F'(t) = -5 \times (-\frac{1}{5})e^{-t/5} = e^{-t/5}$. ✓ The factor -5 is $1/a$ with $a = -\frac{1}{5}$; writing $e^{-t/5}$ as its own antiderivative, which is the usual slip, would give a derivative $-\frac{1}{5}e^{-t/5}$ instead.



Part (a) read as an area. The shaded region is a trapezium of width 3 whose parallel sides have lengths $f(0) = 1$ and $f(3) = 7$, so its area is $3 \times \frac{1+7}{2} = 12$ — the same number the antiderivative gives, obtained without any calculus at all.

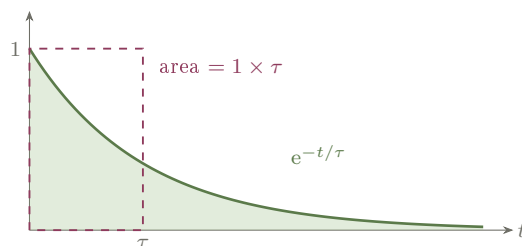
Exercise 9 (a) Show that $\int_0^\infty e^{-t/\tau} dt = \tau$. (b) A rate of responding is $r(t) = 4$ per minute for 10 minutes: write the total as an integral and compute it. (c) In one sentence, how are a derivative and an integral related?

- (a) An antiderivative of $e^{-t/\tau}$ is $F(t) = -\tau e^{-t/\tau}$; differentiate it to check. Since $\tau > 0$, $e^{-t/\tau} \rightarrow 0$ as $t \rightarrow \infty$, so $F(t) \rightarrow 0$, while $F(0) = -\tau$. Hence

$$\int_0^\infty e^{-t/\tau} dt = 0 - (-\tau) = \tau.$$

The condition $\tau > 0$ is what makes the upper limit vanish; for $\tau < 0$ the integral does not converge at all.

- (b) The rate is constant, so $\int_0^{10} 4 dt = [4t]_0^{10} = 40$ responses. A constant rate integrates to rate $\times$ duration, which is the area of a rectangle — the general statement “a rate integrates to a total” in its simplest case.
- (c) They are inverse operations: differentiating a quantity gives its rate, and integrating a rate gives back the accumulated quantity, up to the one constant that the initial value fixes.



Part (a) as a picture. The green region continues to the right without end, yet its total area is finite and equals the area of the dashed rectangle of height 1 and width τ . The time constant therefore has a second meaning: it is the area under the decay.

Functions of several variables

Exercise 10 *Partial derivatives. Compute $\partial f/\partial x$ and $\partial f/\partial y$ for (a) $f = x^2 + 3xy + y$; (b) $f = x^2 + 4y^2$; (c) $f = 5x + 2y - 7$; (d) $f = xe^{-y}$.*

In each case one letter is the variable and the other is held at a fixed value, so it behaves exactly like a number: it survives as a multiplicative factor, and it differentiates to zero when it stands alone.

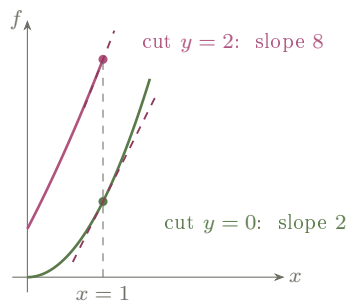
- (a) $\frac{\partial f}{\partial x} = 2x + 3y$ — in the term $3xy$ the factor $3y$ is a constant. $\frac{\partial f}{\partial y} = 3x + 1$ — here x^2 is the constant and differentiates to 0.
- (b) $\frac{\partial f}{\partial x} = 2x$, $\frac{\partial f}{\partial y} = 8y$.
- (c) $\frac{\partial f}{\partial x} = 5$, $\frac{\partial f}{\partial y} = 2$. Both are constants: the function is linear, so its slope in each direction is the same at every point. This is the exception, not the rule — and it is precisely what fails in (a).
- (d) $\frac{\partial f}{\partial x} = e^{-y}$ — in x , the factor e^{-y} is a constant. $\frac{\partial f}{\partial y} = -xe^{-y}$ — in y , the factor x is a constant and the exponential rule applies with $a = -1$.

Watch out — the usual mistake

The frequent error in (a) is $\partial(3xy)/\partial x = 3$. The y does not disappear: it is a constant, and a constant factor is kept, exactly as the 5 was kept in $5e^{-2t}$.

Exercise 11 (a) For $f(x, y) = x^2 + 3xy + y$, evaluate $\partial f/\partial x$ at $(1, 0)$ and at $(1, 2)$. (b) Why are the two answers different?

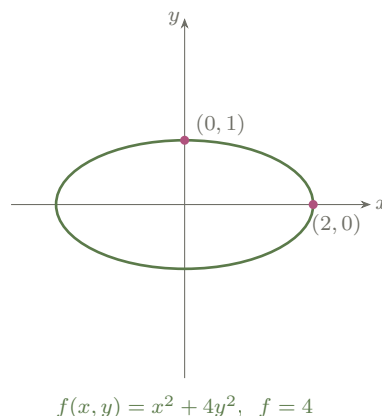
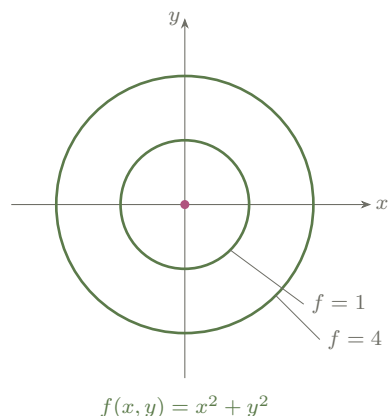
- (a) Differentiate first, substitute afterwards. From the previous exercise, $\partial f/\partial x = 2x + 3y$, so at $(1, 0)$ it is 2 and at $(1, 2)$ it is $2 + 6 = 8$.
- (b) Because the slope in the x direction contains y . Fixing y turns f into a function of x alone — at $y = 0$ it is x^2 , at $y = 2$ it is $x^2 + 6x + 2$ — and these two curves have different slopes above $x = 1$. The two variables *interact*, through the term $3xy$, and that interaction is the one genuinely new feature of functions of several variables. Where it is absent, as in Exercise 10(c), the partial derivatives are constant.



Holding y fixed cuts the surface along a curve. The two cuts $y = 0$ and $y = 2$ are different curves, and above $x = 1$ they have different slopes: 2 and 8. The dashed wine lines are those two tangents. (The vertical scale of the pink cut has been compressed so that both fit on one picture.)

Exercise 12 (a) Sketch the contour lines $f = 1$ and $f = 4$ of $f(x, y) = x^2 + y^2$; what shape are they? (b) Sketch the contour $f = 4$ of $f(x, y) = x^2 + 4y^2$. (c) On a contour line, does f change? Where is the surface steepest?

- (a) $x^2 + y^2 = c$ is the set of points at distance $\sqrt{c}$ from the origin, so the contours are **circles** centred at the origin, of radius 1 and 2.
- (b) $x^2 + 4y^2 = 4$; dividing by 4 gives $\frac{x^2}{4} + y^2 = 1$, an **ellipse** through $(\pm 2, 0)$ and $(0, \pm 1)$ — twice as wide as it is tall, because the coefficient 4 makes f grow four times faster in y than in x .
- (c) No: f is constant along a contour, by the definition of a contour. The surface is steepest where the contour lines are **closest together**, since a small step there crosses from one level to the next.



Left, part (a): the contours of $x^2 + y^2$ are circles, and they crowd together as one moves outwards — the surface gets steeper. Right, part (b): the same value $f = 4$ gives an ellipse once the y term is weighted by 4; along y the surface climbs faster, so the contour is reached sooner.

Exercise 13 Vector-valued functions. (a) $\mathbf{x}(t) = (2t, 3 - t)$: compute $\dot{\mathbf{x}}$, the speed, and the shape of the path. (b) $\mathbf{x}(t) = (t, t^2)$: compute $\dot{\mathbf{x}}$ and the velocity at $t = 0$ and $t = 1$. (c) $\mathbf{x}(t) = (t, e^{-t})$: compute $\dot{\mathbf{x}}$ and the velocity at $t = 0$.

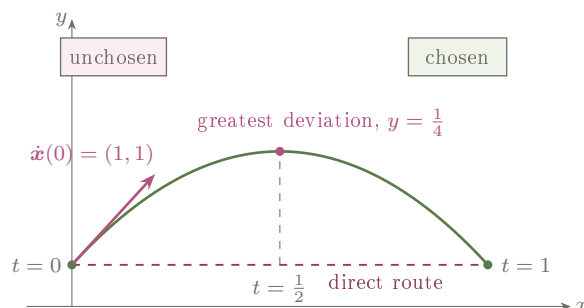
- (a) Differentiate each component separately: $\dot{\mathbf{x}}(t) = (2, -1)$, the same vector at every t . Its

length is $\|\dot{\mathbf{x}}\| = \sqrt{4+1} = \sqrt{5}$, so the speed is constant. A velocity that never changes direction traces a **straight line**; eliminating t confirms it, since $x = 2t$ and $y = 3 - t$ give $y = 3 - x/2$.

- (b) $\dot{\mathbf{x}}(t) = (1, 2t)$. At $t = 0$ this is $(1, 0)$: horizontal. At $t = 1$ it is $(1, 2)$: still moving right, but now climbing twice as fast as it advances. The speed $\sqrt{1+4t^2}$ increases with t .
- (c) $\dot{\mathbf{x}}(t) = (1, -e^{-t})$, so at $t = 0$ the velocity is $(1, -1)$: moving right and downwards at 45° . As t grows the second component tends to 0 and the motion becomes horizontal.

Exercise 14 A mouse-tracking trajectory is $\mathbf{x}(t) = (t, t - t^2)$ for $t \in [0, 1]$. (a) Where is the participant at $t = 0$ and $t = 1$? (b) Compute $\dot{\mathbf{x}}(t)$ and the velocity at $t = 0$. (c) At which t is y largest, and what does that moment represent?

- (a) At $t = 0$: $\mathbf{x} = (0, 0)$, the starting position. At $t = 1$: $\mathbf{x} = (1, 0)$, which is therefore where the chosen button sits.
- (b) $\dot{\mathbf{x}}(t) = (1, 1 - 2t)$. At $t = 0$ this is $(1, 1)$: the cursor leaves at 45° above the straight line joining the start to the chosen button.
- (c) $y(t) = t - t^2$ is a function of one variable, so use the derivative: $\dot{y} = 1 - 2t$ vanishes at $t = \frac{1}{2}$, and $\dot{y} > 0$ before, $\dot{y} < 0$ after — a maximum, with $y(\frac{1}{2}) = \frac{1}{4}$. It is the moment of greatest deviation from the direct route, that is, the moment at which the pull of the unchosen option is strongest, just before the participant commits.



The trajectory of Exercise 14. Three things are shown and kept distinct: the recorded path (green), the marked moments and the initial velocity (pink), and the direct route that the participant did *not* take (dashed wine). The cursor sets off at 45° towards the unchosen side, reaches its greatest deviation at $t = \frac{1}{2}$, and only then turns towards the button it selects.

Differential equations

Exercise 15 For each, does the rate depend on time (integrate) or on the quantity itself (a differential equation)? (a) $\dot{x} = 3t^2$; (b) $\dot{x} = -2x$; (c) $\dot{x} = -2x + 5$; (d) $\dot{x} = e^{-t}$.

The test is purely one of inspection: look at which letter appears on the right-hand side.

- (a) Only t appears: an **integration** problem, $x(t) = t^3 + C$.

- (b) x appears: a **differential equation**. There is nothing to integrate, because the right-hand side is not yet known as a function of t — it depends on the very thing being sought.
- (c) x appears: a **differential equation**, of the form $\dot{x} = ax + b$.
- (d) Only t appears: an **integration** problem, $x(t) = -e^{-t} + C$.

Exercise 16 (a) Solve $\dot{x} = -2x$ with $x_0 = 5$; where does x go? (b) Solve $\dot{x} = 0.5x$ with $x_0 = 2$; where does x go? (c) Check both by differentiating.

- (a) The equation is $\dot{x} = ax$ with $a = -2$, whose solutions are $x(t) = x_0 e^{at}$. Hence $x(t) = 5e^{-2t}$. Since $a < 0$, x decreases and tends to 0, without reaching it.
- (b) Same form with $a = 0.5 > 0$: $x(t) = 2e^{0.5t}$, which **increases without bound**. The formula is identical; only the sign of a has changed, and with it the entire qualitative behaviour.
- (c) $\frac{d}{dt}(5e^{-2t}) = -10e^{-2t} = -2x$. ✓ $\frac{d}{dt}(2e^{0.5t}) = e^{0.5t} = 0.5 \times 2e^{0.5t} = 0.5x$. ✓

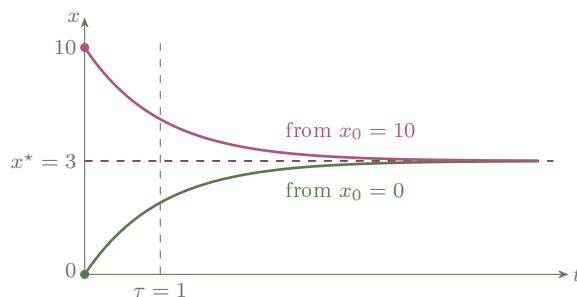
Exercise 17 For $\dot{x} = ax + b$, with $x(t) = x^* + (x_0 - x^*)e^{at}$ and $x^* = -b/a$. (a) $\dot{x} = -x + 3$: find x^* and τ , write $x(t)$ from $x_0 = 0$; stable? (b) The same equation from $x_0 = 10$. (c) Check that $x(t) = 3 + Ce^{-t}$ solves $\dot{x} = -(x - 3)$ for every C ; why is an initial condition needed?

- (a) Setting the rate to zero: $-x + 3 = 0$ gives $x^* = 3$. Here $a = -1$, so $\tau = -1/a = 1$, and $a < 0$, so the fixed point is **stable**. With $x_0 = 0$,

$$x(t) = 3 + (0 - 3)e^{-t} = 3 - 3e^{-t},$$

which increases from 0 towards 3.

- (b) Same x^* and same τ ; only the initial gap changes: $x(t) = 3 + 7e^{-t}$, which decreases from 10 towards 3. The two solutions approach the same value from opposite sides, at the same rate: what decays exponentially is the *gap* $x - x^*$, not x itself.
- (c) Differentiating, $\dot{x} = -Ce^{-t}$; and the right-hand side is $-(x - 3) = -(3 + Ce^{-t} - 3) = -Ce^{-t}$. The two agree for every C . ✓ The equation alone therefore has infinitely many solutions, one for each C ; they differ only in where they start. Prescribing $x(0)$ selects one of them, since $x(0) = 3 + C$ fixes C .



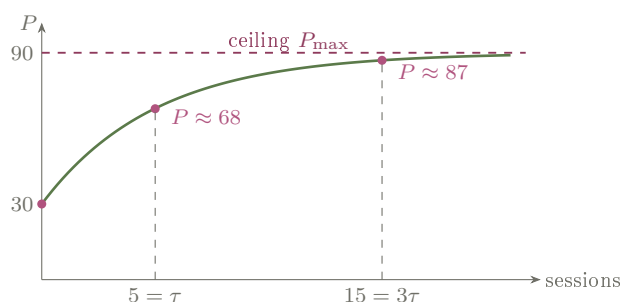
Parts (a) and (b): one equation, two initial conditions. Both curves approach the same steady state, and after one time constant each has closed 63% of its own initial gap to it. The vertical scales of the two gaps differ, but the rate at which they shrink does not: that rate is τ , and it belongs to the equation, not to the starting point.

Exercise 18 Someone proposes $-x^2/2 + 3x$ as the solution of $\dot{x} = -(x-3)$. (a) What operation did they perform? (b) Give the one-line reason it cannot be a solution.

- (a) They antidifferentiated the right-hand side with respect to x : an antiderivative of $-(x-3) = -x+3$ is indeed $-x^2/2 + 3x$.
- (b) Because it is a function of x , whereas the solution of a differential equation must be a function of t : the two objects are not even of the same kind, so no computation is needed to reject it. Integration is the right move only when the rate depends on time alone, as in $\dot{x} = f(t)$; here it depends on x , so the unknown stands on both sides and there is nothing yet to integrate.

Exercise 19 Performance obeys $\dot{P} = k(P_{\max} - P)$, with $P_{\max} = 90$, $k = 0.2$ per session, $P_0 = 30$. (a) Give P^* and τ . (b) Write $P(t)$, compute $P(5)$ and $P(15)$. (c) How many sessions close 95% of the gap? (d) What do the forgetting and learning curves have in common?

- (a) Expand the bracket to reach the standard form: $\dot{P} = -kP + kP_{\max}$, so $a = -k = -0.2$ and $b = kP_{\max} = 18$. Then $P^* = -b/a = P_{\max} = \mathbf{90}$ and $\tau = -1/a = 1/k = \mathbf{5}$ sessions. Note $\tau = 1/k$, not k .
- (b) $P(t) = P^* + (P_0 - P^*)e^{-t/\tau} = 90 - 60e^{-t/5}$.
 $P(5) = 90 - 60(0.37) \approx \mathbf{68}$ (one time constant), $P(15) = 90 - 60(0.05) \approx \mathbf{87}$ (three).
- (c) Closing 95% of the gap means $e^{-t/\tau} = 0.05$, that is $t \approx 3\tau$: about **15** sessions. The same three numbers as everywhere else in this session.
- (d) Both say that the rate of change is proportional to the distance still to be travelled — to zero for forgetting, to the ceiling for learning. They are the same equation $\dot{x} = ax + b$ with $a < 0$, run in opposite directions.



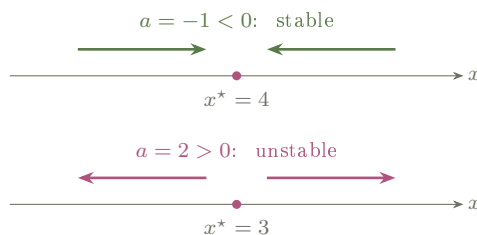
The learning curve of Exercise 19. The initial gap is $90 - 30 = 60$; after one time constant 37% of it remains, after three about 5%. The curve never touches the dashed ceiling, which is why (c) asks for 95% of the gap rather than all of it.

Exercise 20 Stability without solving. (a) $\dot{x} = -x + 4$: find x^* and say what happens from $x = 10$ and from $x = 0$. (b) $\dot{x} = 2x - 6$: the same; is it stable? (c) Complete: the fixed point is stable when a is _____ and unstable when a is _____.

- (a) $-x + 4 = 0$ gives $x^* = 4$. There is no need to solve the equation: evaluate the rate on each

side. At $x = 10$ the rate is $-6 < 0$, so x decreases towards 4; at $x = 0$ the rate is $+4 > 0$, so x increases towards 4. Since $a = -1 < 0$ the fixed point is **stable**, and both motions confirm it.

- (b) $2x - 6 = 0$ gives $x^* = 3$, and $a = 2 > 0$: **unstable**. At $x = 3.1$ the rate is $+0.2$, so x runs away upwards; at $x = 2.9$ it is -0.2 and x runs away downwards. A fixed point of this kind is a threshold to be crossed, not a destination to be reached.
- (c) Stable when a is **negative**; unstable when a is **positive**. The case $a = 0$ is excluded: the formula $x^* = -b/a$ then divides by zero, and indeed $\dot{x} = b$ has no steady state at all.



The two situations of Exercise 20, drawn without solving anything. Each arrow shows the direction in which x moves, read off the sign of the rate. Above, the arrows point towards the fixed point; below, away from it. The sign of a is the sign of the slope of the rate at x^* , and it alone decides which of the two pictures applies.

Key idea

The whole of this session is three statements, and every exercise above is one of them in disguise.

- e^{at} increases when $a > 0$ and decreases when $a < 0$; when $a < 0$ the number $\tau = -1/a$ is the time constant, and each step of τ multiplies what remains by 0.37.
- Differentiating gives a rate; integrating a rate gives back the total, up to the one constant an initial value fixes. The sign of a derivative says increasing, decreasing or stationary, and that is often all one needs.
- $\dot{x} = ax + b$ has steady state $x^* = -b/a$, and the distance from x to x^* is multiplied by e^{at} . It therefore shrinks exactly when $a < 0$, which is what “stable” means.